

Experimental setup

NetLogo · BehaviorSpace

BA benchmark network

$N = 1,000$

Initial pollution

$p_0 = 10\%, 30\%, 50\%$

Parameter sweeps

10 runs $\times$ 50 ticks

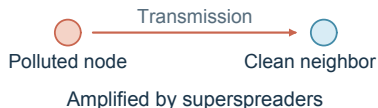
01 Pollution diffusion

Network-mediated transmission



● Clean ● Polluted
⊙ Ring: superspreader
Top 1% of nodes by degree

Exposure along network edges



02 Governance intervention

One fixed regime per run

⊘ **No governance**
No intervention

🏠 **Legal governance**

Coverage L per round
Response interval τ_L
Decision accuracy α_L

🖥️ **Platform governance**

Audit coverage n_p per tick
Every-tick moderation
Classification accuracy α

Both governance actors

Random or degree-targeted

03 Outcome evaluation

Run-level outcomes

PRIMARY

Mean effectiveness

$\overline{E}$ Mean clean-node share
over ticks 1–50

Cumulative collateral exposure

$O(50)$
Share of agents ever exposed
to wrongful intervention

SUPPLEMENTARY

Removal precision $PR(50)$

Net effectiveness $E_{\text{net}}(t)$

10 independent runs / condition
Condition means + 95% CIs

Next tick · repeat for 50 ticks

04 Robustness analysis

Re-run benchmark conditions

Network topology



BA



ER



WS

Network size

$N = 1,000$

$N = 5,000$